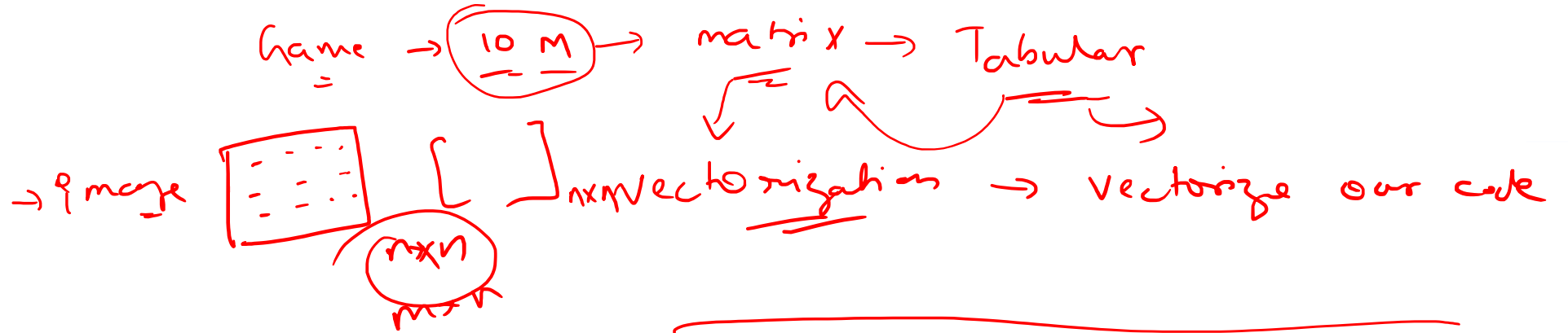




Singular Value Decomposition

Arpit Jain, PhD

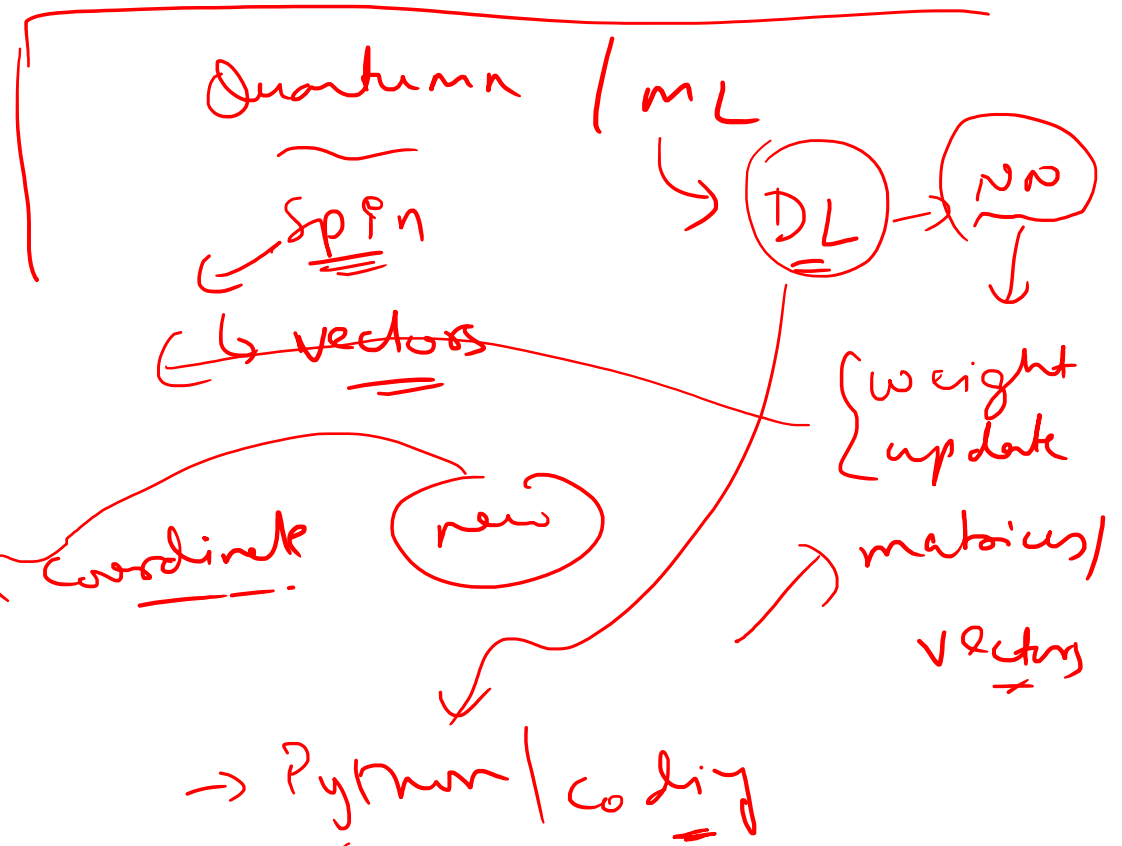
Recap



Operation

- Matrices
- Vectors
- Basis and unit vectors
- Linear combination of vectors
- Linear Transformations $\rightarrow$ Phys.
- $\rightarrow$ • Change of Basis $\rightarrow$ reference $\rightarrow$ Coordinate
- Eigenvectors and Eigenvalues
- Inner Product

$\rightarrow$ Scalars



Eigendecomposition

Suppose we have a symmetric matrix A of $(n \times n)$ dimensions. This matrix can be represented as:

$$A = PDP^{-1}$$

Where,

→ D is an $(n \times n)$ diagonal matrix comprised of eigen values of A .

The columns of matrix P are the n linearly independent eigenvectors of A corresponding to the eigenvalues in D respectively.

Hence,

$$A = \underbrace{\begin{bmatrix} | & | & & | \\ u_1 & u_2 & \dots & u_n \\ | & | & & | \end{bmatrix}}_P \begin{bmatrix} \lambda_1 & 0 & \dots & 0 \\ 0 & \lambda_2 & \dots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \dots & \lambda_n \end{bmatrix} \begin{bmatrix} | & | & & | \\ u_1 & u_2 & \dots & u_n \\ | & | & & | \end{bmatrix}^{-1}$$

Symmetric

→ Square matrix

$$A = A^T$$

10/15/...
Dataset $n \times m$
242

Note: eigendecomposition is important when we want to analyze a given problem statement on form of eigenvalues and eigenvectors. Mostly this concept or an extension (SVD) of it is used in Dimensionality reduction.

SVD

PCA

Int.

game

10m

2475

10m

Huge 2475

Great

15.9 billion = 2475 years

Eigendecomposition

E.X.

$$A = \begin{bmatrix} 3 & 1 \\ 1 & 2 \end{bmatrix} \Rightarrow$$

$$A^T = \begin{bmatrix} 3 & 1 \\ 1 & 2 \end{bmatrix}$$

$$A = A^T$$

$$A = P D P^{-1}$$

$\downarrow \quad \downarrow \quad \downarrow$
 eigen vectors -
 eigen values -
 eigen vectors -

$$A = [u_1 \ u_2] [D] [u_1 \ u_2]^{-1}$$

$$\lambda_1 = 3.618 \quad \lambda_2 = 1.382$$

$$u_1 = \begin{bmatrix} 0.8507 \\ 0.5257 \end{bmatrix}$$

$$u_2 = \begin{bmatrix} -0.5257 \\ 0.8507 \end{bmatrix}$$

$$[u_1 \ u_2] = P$$

$$D = \begin{bmatrix} 3.618 & 0 \\ 0 & 1.382 \end{bmatrix}$$

$$P = \begin{bmatrix} 0.8507 & -0.5257 \\ 0.5257 & 0.8507 \end{bmatrix}$$

Singular Value Decomposition (SVD)

→ Dimensionality reduction

The eigendecomposition was applicable only to the symmetric matrices. SVD is a generalized form of eigendecomposition which can be applied to non-symmetric or even rectangular matrices.

The SVD of a matrix is a factorization of that matrix into three matrices. This decomposition conveys some important geometrical and algebraic properties and provides insights about the problem statement/transformations.

Mathematically:

$$\begin{bmatrix} \lambda_1 & 0 \\ 0 & \lambda_2 \\ 0 & 0 \end{bmatrix}$$

Where,

M is a given $m \times n$ matrix.

U is an orthogonal matrix

Σ is a diagonal matrix

V^T is an orthogonal matrix

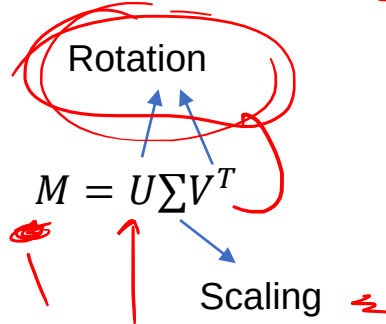
Physically SVD can be visualised as:

$$M = U \Sigma V^T$$

$$m = u \Sigma v^T$$

Left Singular matrix

Right Singular matrix
diagonal matrix



From Scale
to Scale

$$A = U \Sigma V^T$$

$U \rightarrow$ orthogonal

$$A^T \cdot A = (V \Sigma^T \underbrace{U^T U}_{I}) \cdot (U \Sigma V^T)$$

$$= V \underbrace{\Sigma^T \Sigma}_{\text{diagonal matrix}} V^T$$

$\rightarrow$ eigenmatrix

$A^T \cdot A \rightarrow V$
eigenvektor

$V \rightarrow$ orthogonal

$$A \cdot A^T = (U \Sigma \underbrace{V^T V}_{I}) \cdot (V \Sigma^T U^T)$$

$$= U \Sigma \Sigma^T U^T$$

$\rightarrow$ eigenvektor

$A \cdot A^T \rightarrow U$
eigenvektor

References

$$A = \begin{bmatrix} 3 & 1 \\ 0 & 2 \end{bmatrix} \quad \text{A} \rightarrow \underbrace{\begin{bmatrix} 3 & 1 & 0 \\ 2 & 2 & -1 \end{bmatrix}}_{\text{SVD}}$$

- <https://atozmath.com/example/MatrixEv.aspx?q=svd&q1=E1>
- <https://gregorygundersen.com/blog/2018/12/10/svd/>
- [https://web.mit.edu/be.400/www/SVD/Singular Value Decomposition.htm](https://web.mit.edu/be.400/www/SVD/Singular%20Value%20Decomposition.htm)
- http://people.whitman.edu/~hundledr/courses/M240S16/M240/Th_eSVD.pdf

ML/DL

eigenvalues

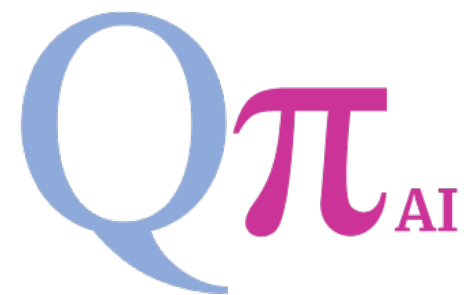
python
✓ internally

Data
Sci

or
Dirig

Code ← Dirig
↑
SVD
→

Comput-d



Thank you!